[Thm. Let $f:[a,b]\to\mathbb{R}$ be a Riemann-integrable, bounded map.
Then, $F:[a,b]\to\mathbb{R}; x\mapsto \int_a^x f(t)\,dt$

is continuous uniformly. Moreover, if f is continuous at $x_0\in[a,b]$, then the derivative

$$F'(x_0)=f(x_0) \quad \text{exists}.$$

Proof. Let $M>0$ be a bound for $|f|$. Then, given $x,y\in[a,b]$,

$$|F(x)-F(y)|=\left|\int_y^x f(t)\,dt\right| \le \int_y^x |f(t)|\,dt \le M\cdot|x-y|.$$

Thus, given $\varepsilon>0$, we can take $\delta=\frac{\varepsilon}{M}$, to uniformly continuity.

Now, let $x_0\in[a,b]$ s.t f is continuous at x_0.
Given $\varepsilon>0$, take $\delta>0$ such that

$$|x-x_0|<\delta \Rightarrow |f(x)-f(x_0)|<\varepsilon$$

Now, if $0<|x-x|<\delta$, then

$$\left|\frac{F(x)-F(x_0)}{x-x_0}-f(x_0)\right| = \left|\frac{1}{x-x_0}\cdot\left(\int_{x_0}^x f(t)\,dt\right)-f(x_0)\right|$$

$$= \left|\frac{1}{x-x_0}\cdot\int_{x_0}^x f(t)-f(x_0)\,dt\right| < \varepsilon.$$

[Thm. If $f:[a,b]\to\mathbb{R}$ is Riemann-integrable and $F'=f$, then

$$\int_a^b f(t)\,dt = F(b)-F(a).$$

Proof. Let $P=\{x_0,\cdots,x_n\}$ be an arbitrary partition on $[a,b]$.
Then, by the mean-value Theorem, for each $1\le i\le n$,

$$F(x_i)-F(x_{i-1})=F'(t_i)\cdot(x_i-x_{i-1}) \text{ for some } t_i\in(x_{i-1},x_i).$$

So, $F(b)-F(a)=\sum_{k=1}^n \left(F(x_i)-F(x_{i-1})\right)=\sum_{k=1}^n f(t_i)\cdot(x_i-x_{i-1})$.

Now, since $L(f,P)\le \sum_{k=1}^n f(t_i)\cdot(x_i-x_{i-1})\le U(f,P)$,

the Squeeze Theorem gives the result